

# Ziquan Wei

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## EDUCATION

- University of North Carolina at Chapel Hill, USA *PhD in Computer Science, Aug 2023 - May 2028*
- Huazhong University of Science and Technology, China *Master in Biomedical Engineering, Sept 2019 - June 2022*
- Yunnan Normal University, China *Bachelor in Computer Science, Sept 2015 - June 2019*

## RESEARCH INTEREST

**Research scientist (intern) in generative models and post-training.** I build conditional diffusion and VQ-VAE models over image, mesh, graph, and time-series data, pretrain a 1.2B multitask model and a Mixture-of-Experts encoder, and post-train language models with reward modeling and PPO.

## RESEARCH EXPERIENCE

- University of North Carolina at Chapel Hill  *Research Assistant, Aug 2023 - Current*
  - **Conditional Generative Models for Digital Twins:** Generate 4D cardiac mesh sequences over the full cycle from the surface ECG [5]. Generate counterfactual disease trajectories with conditional latent diffusion over tokenized event sequences, factorized as  $P(\text{Future}|\text{Biomarker}) \cdot P(\text{Biomarker}|\text{History})$  with imaging biomarkers as mediators, over ~500K subjects and 1,000+ diseases [3][16]. Design SPD-VQVAE, a VQ-VAE whose decoder emits a Cholesky factor so every generated covariance matrix is positive definite by theory.
  - **Foundation Model Pretraining:** 1.2B-parameter multitask pretraining [1] and a Mixture-of-Experts encoder that holds accuracy under site and scanner shift [14].
  - **Geometric Deep Learning:** Graph transformers over pathway structure rather than pairwise edges [18][4]; a Riemannian manifold mixer on the scattering transform [20].
  - **LLM Agents:** Built an 11-role agent graph over an MCP tool layer with A2A messaging on local weights, evaluated on a 500-query benchmark at 93.6% task success; and an agent pipeline for neuroimage processing and quality control [10].
- EGRA.ai  *Research Intern, May 2026 - Aug 2026*
  - **Generative EEG Twin:** Built a text-conditioned diffusion model that generates per-fixation EEG for a reader from frozen Qwen3-Embedding features and demographics, trained over 60 subjects and 416K fixations.
  - **Reward Modeling and RL Post-Training:** Fed generated EEG through a frozen encoder into Bradley-Terry reward models on four base LMs with LoRA, as input feature and as privileged supervision, across 174 5-fold CV cells; post-trained a Llama-3.2-1B policy with PPO against the learned reward, evaluated on held-out RewardBench v1/v2 and 3-seed win rate.
- University of North Carolina at Chapel Hill *Predoctoral Fellow, Aug 2022 - Aug 2023*
  - **3D Nuclei Instance Segmentation:** Built a 2D-to-3D algorithm for anisotropic whole-brain lightsheet volumes, segmenting in the isotropic X-Y plane and tracking instances through the Z stack, at over 90% precision and recall and 15 hr per brain, one third of CellPose v2, over 30 to 50 million nuclei; shipped as CellPheno over a 118-volume cohort [24][6][11].
- Huazhong University of Science and Technology *Graduate Researcher (MSc), Sept 2019 - June 2022*
  - **High-Throughput Gigapixel Inference:** Built YOLCO, cutting forward passes per slide by two orders of magnitude and a gigapixel slide from 22.5 minutes to 55.9s in the deployed C++ release, at 0.872 AUC over 2,019 slides from four scanners [27]; and a detector under distribution shift [26].
- Nanyang Technological University, Singapore *Research Intern, Jun 2021 - Sept 2021*
  - **Audio-Visual Emotion Recognition:** Built a spatial-temporal network, 2D-CNN plus TCN for video and parallel TCNs for audio, joined by leader-follower attentive fusion. A 300-frame window against 36 to 48 for prior work gave CCC 0.463 valence and 0.492 arousal on Aff-Wild2, against baseline 0.200 and 0.190 [28].
- Yunnan Normal University *Research Assistant, Sept 2016 - June 2019*
  - **Non-Rigid Registration:** Multi-temporal registration of UAV imagery [32]; non-rigid point set and image registration with finite mixture models under structure and curvature constraints [31][30].

## SKILLS

- **Generative Models:** Conditional latent diffusion, VQ-VAE, digital twins, counterfactual sampling
- **RL Post-Training:** Bradley-Terry reward modeling, PPO with LoRA, RewardBench and best-of-N evaluation
- **LLM Agents & NLP:** Multi-agent orchestration, MCP tools, language modeling, Mixture-of-Experts pretraining
- **Computer Vision:** 3D instance segmentation, object detection, non-rigid registration, 4D mesh generation
- **Engineering & Data:** Python, PyTorch, C++, distributed training, HPC admin (20 GPUs), Google Cloud Platform

- [1] Wei Z., Dan T., Wu G. **Large Connectome Model: An fMRI Foundation Model of Brain Connectomes Empowered by Brain-Environment Interaction in Multitask Learning Landscape.** *AAAI* '26.
- [2] Wei Z., Wu G. **Teravoxel Microscopy Image Analysis for Neurological Diseases.** *ARBME* '26 (IF=11.5).
- [3] Wei Z., Dan T., Wu G. **Marrying Generative Model of Healthcare Events with Digital Twin of Social Determinants of Health for Disease Reasoning.** *ICML* '26.
- [4] Wei Z., Dan T., Ding J., Laurienti P.J., Wu G. **NeuroDetour: A Neural Pathway Transformer for Uncovering Structural-Functional Coupling Mechanisms in Human Connectome.** *MedIA* '26 (IF=14.0).
- [5] Wei Z., Dan T., Wu G. **From Heartbeat to Cardiochoreography: A Mechanics Foundation Model for Individualized 4D Cardiac Motion Generation Conditioned on Electrophysiology.** *NeurIPS* '26.
- [6] Wei Z., Curtin I., Kyere F.A., et al. **CellPheno: A High-throughput Computational Platform for Quantifying Cellular Resolution Whole Brain Microscopy Images.** *bioRxiv* '26.
- [7] Ding J., Dan T., Wei Z., Laurienti P.J., Wu G. **Scanning the Horizon of Replicability in Neuroscience: A Recipe of Developing Replicable Deep Models for Functional Neuroimages.** *IEEE TBME* '26 (IF=4.5).
- [8] Wen T., Wei Z., Dan T., et al. **Image Imputation for Real-World Defective Microscopy Images of Mouse Brains.** *ISBI* '26.
- [9] Ding J., Dan T., Wei Z., Laurienti P.J., Wu G. **Machine learning on dynamic functional connectivity: Promise, pitfalls, and interpretations.** *Inf. Sci.* '26 (IF=6.0).
- [10] Chen Y., Li Y., Tong J., ..., Wei Z., et al. **NeuroPilot: An Agent-Driven Smart Pipeline for Processing, Quality Control, and Managing Neuroimages.** *arXiv* '26.
- [11] Kyere F.†, Curtin I.†, Wei Z.†, et al. **Chd8 haploinsufficiency leads to molecular layer heterotopias and age-dependent cortical expansion.** *bioRxiv* '26.
- [12] Wei Z., Dan T., Ding J., Wu G. **Efficient Graph Representation Learning by Non-Local Information Exchange.** *Electronics* '25 (IF=2.9).
- [13] Wei Z., Dan T., Wu G. **Brain-Environment Cross-Attention (BECA) Meta-matching: A New Perspective of Brain Connectome Zero-Shot Learning.** *MICCAI* '25 (Early accept (9%)).
- [14] Wei Z., Dan T., Chen T., Wu G. **BrainMoE: Cognition Joint Embedding via Mixture-of-Expert Towards Robust Brain Foundation Model.** *NeurIPS* '25.
- [15] Cho H., Wei Z., Lee S., et al. **Conditional Diffusion Model using Ordinal Regression for Longitudinal Neurodegenerative Data Generation.** *Alz. & Dem.* '25 (IF=12.8).
- [16] Cho H., Wei Z., Lee S., et al. **Conditional Diffusion with Ordinal Regression: Longitudinal Data Generation for Neurodegenerative Disease Studies.** *ICLR* '25 (Spotlight poster (3.26%)).
- [17] Wei Z., Dan T., Ding J., Laurienti P., Wu G. **Representing Functional Connectivity with Structural Detour: A New Perspective to Decipher Structure-Function Coupling Mechanism.** *MICCAI* '24.
- [18] Wei Z., Dan T., Ding J., Wu G. **NeuroPath: A Neural Pathway Transformer for Joining the Dots of Human Connectomes.** *NeurIPS* '24.
- [19] Wei Z., Wu G. **Non-local Exchange: Introduce Non-locality via Graph Re-wiring to Graph Neural Networks.** *NeurIPS-W* '24.
- [20] Dan T., Wei Z., Kim W.H., Wu G. **Exploring the Enigma of Neural Dynamics Through A Scattering-Transform Mixer Landscape for Riemannian Manifold.** *ICML* '24.
- [21] Ding J., Dan T., Wei Z., Laurienti P., Wu G. **A Wasserstein Recipe for Replicable Machine Learning on Functional Neuroimages.** *MICCAI* '24.
- [22] Wu Y., Liu Y., Li Y., ..., Wei Z., et al. **Symmetry engineering in 2D bioelectronics facilitating augmented biosensing interfaces.** *PNAS* '24 (IF=9.5).
- [23] Wei Z., Dan T., Ding J., et al. **High Throughput Deep Model of 3D Nucleus Instance Segmentation by Stereo Stitching Contextual Gaps.** *ISBI* '23.
- [24] Wei Z., Dan T., Ding J., Dere M., Wu G. **A General Stitching Solution for Whole-Brain 3D Nuclei Instance Segmentation from Microscopy Images.** *MICCAI* '23 (Early accept (14%)).
- [25] Dan T., Ding J., Wei Z., et al. **Re-think and re-design graph neural networks in spaces of continuous graph diffusion functionals.** *NeurIPS* '23.
- [26] Wei Z., Cheng S., Cai J., et al. **Cervical Glandular Cell Detection from Whole Slide Image with Out-Of-Distribution Data.** *arXiv* '22.
- [27] Wei Z., Cheng S., Hu J., et al. **An Efficient Cervical Whole Slide Image Analysis Framework Based on Multi-scale Semantic and Location Deep Features.** *arXiv* '21.
- [28] Zhang S.†, Ding Y.†, Wei Z.†, Guan C. **Continuous emotion recognition with audio-visual leader-follower attentive fusion.** *ICCVW* '21.
- [29] Cheng S., Liu S., Yu J., ..., Wei Z., et al. **Robust whole slide image analysis for cervical cancer screening using deep learning.** *Nat. Comm.* '21 (IF=18.1).
- [30] Yang Z., Yang Y., Yang K., Wei Z.. **Non-rigid image registration with dynamic Gaussian component density and space curvature preservation.** *IEEE TIP* '18 (IF=13.7).
- [31] Zhang S., Yang K., Yang Y., Luo Y., Wei Z.. **Non-rigid point set registration using dual-feature finite mixture model and global-local structural preservation.** *Pattern Recognit.* '18 (IF=9.1).
- [32] Wei Z., Han Y., Li M., et al. **A small UAV based multi-temporal image registration for dynamic agricultural terrace monitoring.** *Remote Sens.* '17 (IF=4.3).

## HONORS & SERVICE

- **Honors:** Editor of Distinction Award 2025, Scientific Reports (editorial board); Graduate School Transportation Grant, UNC-Chapel Hill; Outstanding Graduate, Huazhong University of Science and Technology
- **Conference Reviewer:** NeurIPS, ICML, ICLR, AAAI, AISTATS, MICCAI
- **Journal Reviewer:** IEEE Transactions on Medical Imaging, NeuroImage, Pattern Recognition

## TALKS

- 2025 Tissue Clearing & Imaging Conference, UNC-Chapel Hill  Speaker, Apr 2025
- **CellPheno:** A High-throughput Computational Platform of Cellular Phenotyping for Real-world Whole Brain Microscopy Images.